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New QSK19 Compact Fuel Pump and Increased Capacity Injector

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document was originally released between 1994 and 2001. It has been added to QSOL for informational purposes

Note : This document supersedes topics 99T5-9, 99T6-9, 00T5-1, and 00T6-10; replace 99T5-9, 99T6-9, 00T5-1, and 00T6-10 with this one.

This document corrects part numbers for the P10 fuel pump (found on page 1) and the OEM interface harness (found on page 5). The correct part numbers are as follows: P10 fuel pump, Part Number 3088415, and OEM interface harness, Part Number 3331775.

This document introduces the new compact fuel pump and the increased capacity trapped volume spill injector for the QSK19.

Note : The product improvements will affect only some new production engines built after December 1998. The present injectors, Part Number 3867761 and Part Number 3867762, and the P10 fuel pump, Part Number 3088415, will continue to be available for service.

Increased Capacity Trapped Volume Spill Injector

The increased capacity trapped volume spill injector, Part Number 4010029, will provide increased fuel-flow capacity for the QSK19 power generation engines and the marine applications that require more power density than the QSK19 industrial applications. This injector will also have common features of the other Quantum™ engine fuel systems, such as the QSK45 and 60. The effect of drain line restriction on start of injection variability, which affects performance and engine durability, will be reduced with the use of the injector. The injector adds robustness to some of the main components with improvements in design and manufacture.

Compact Fuel Pump

The compact fuel pump is about 10 cm [4 in] shorter than the replaced P10 pump. The reduced length permits the use of a two-cylinder air compressor on the QSK19 engines. The compact pump achieves this reduced length by eliminating most of the fuel-wetted components of the P10

pump, which include governor weights, gears, shafts, plungers, sleeves, springs, and shims. The compact pump provides fuel pressure by a precision three-piece gear pump with a floating-gear design for improved efficiency and durability. In the compact pump, the pressure is electronically controlled by an actuator and pressure sensor; this makes it less sensitive to wear and changes in fuel properties by providing more precise control of the pump pressure.

Note : This is a product improvement and is not subject to campaign.

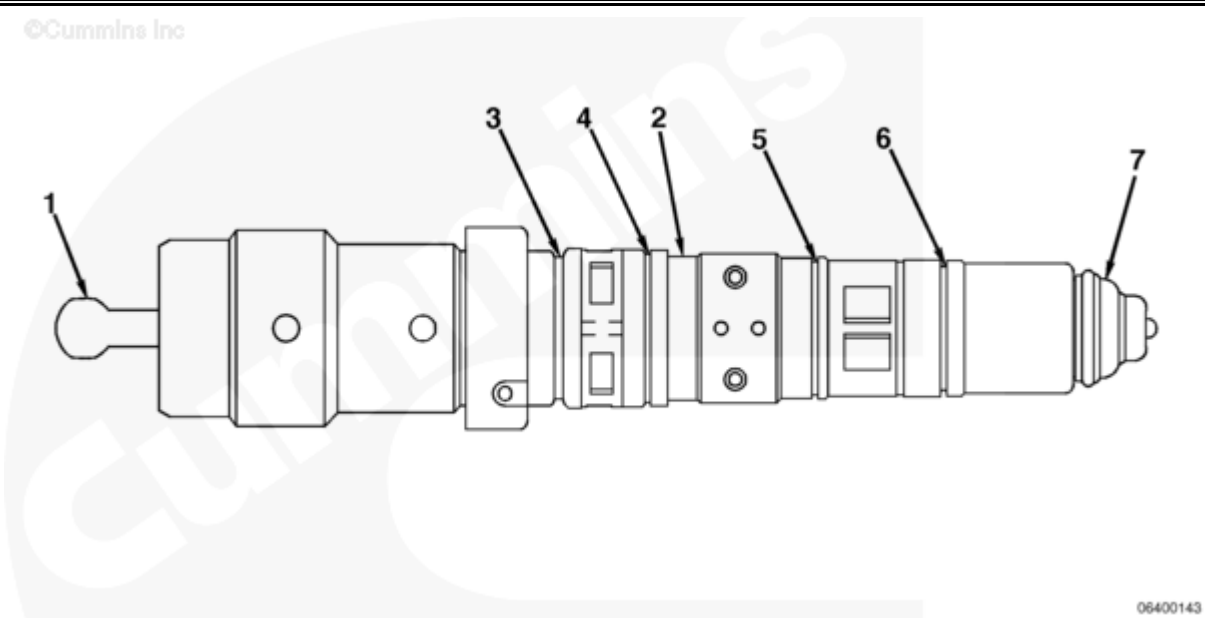
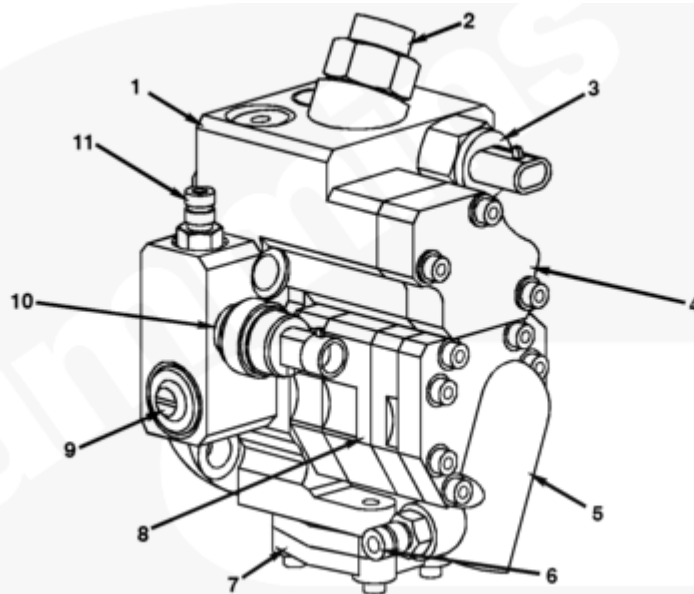


Table 1, New Injector, Part Number 4010029

1. Plunger link	5. Upper rail o-ring
2. Spill ring	6. Lower rail o-ring
3. Upper timing o-ring	7. Copper seal
4. Lower timing o-ring	



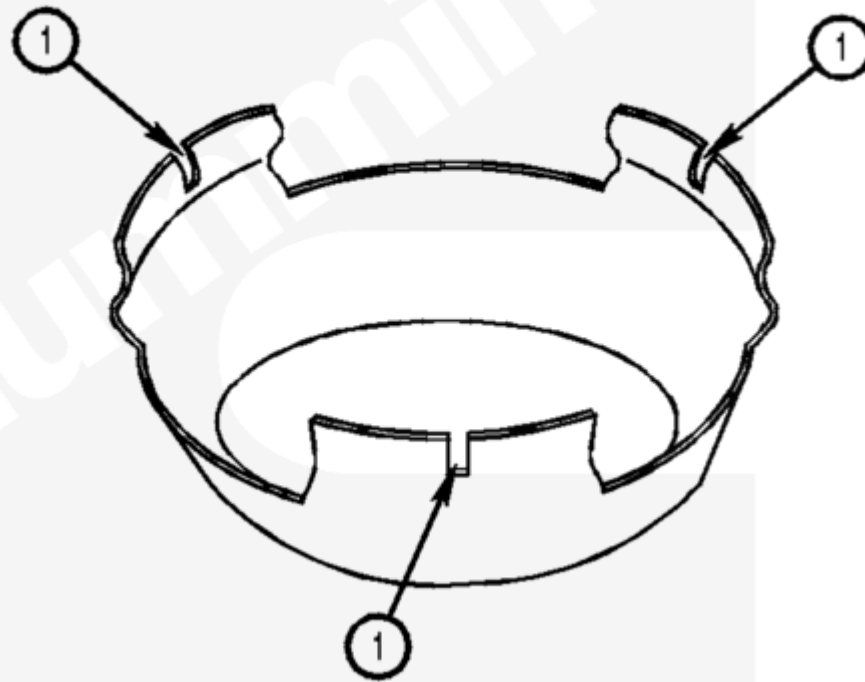
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Table 2, Compact Fuel Pump

1. Fuel pump support	7. Pulsation damper
2. Outlet fitting/check valve	8. Three-piece gear pump
3. Actuator	9. Filter screen
4. Bypass regulator (inside)	10. Pressure sensor
5. Inlet fitting suction head	11. Output pressure Compuchek® fitting
6. Inlet restriction Compuchek® fitting	

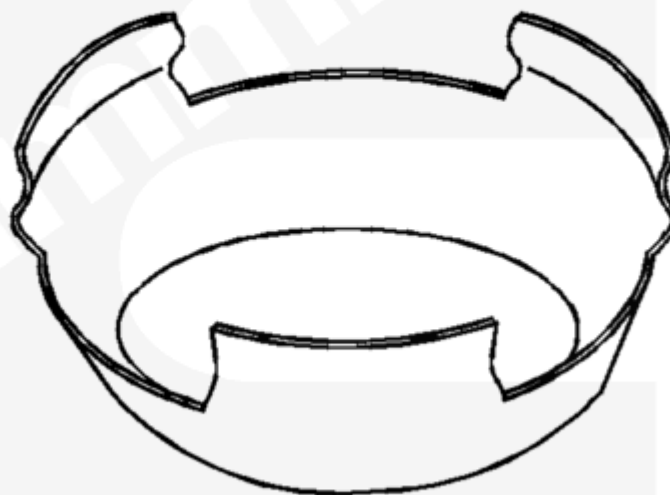
Injector Copper Seal

The new design of the new injector copper seal, Part Number 3867687, has a split in each of the tangs and is **not** interchangeable with the older injector copper seals, Part Number 207244.



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Fig. 1, New Injector Copper Seal, Part Number 3867687, Split Tang (1)



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Fig. 2, Older Injector Copper Seal, Part Number 207244, No Split in Tang

Harnesses and Brackets

The engine harness, OEM harness, ECM, and brackets have been changed as of January 1998. They will continue to be used with the new injector and the compact fuel pump. The part numbers follow:

- Engine harness, Part Number 3331774
- OEM interface harness, Part Number 3331775
- ECM, Part Number 3867743

Calibrations

New fuel system and new electronic fault codes have been established. **The new fault codes are:**

- 118 - Fuel Pump Pressure Sensor Circuit Failed High
- 119 - Fuel Pump Pressure Sensor Circuit Failed Low
- 316 - Fuel Pump Actuator Circuit Error
- 318 - Fuel Pump Fueling Flow Mismatched.

Note : For additional details on fault code descriptions, refer to the Wiring Diagram, Bulletin 3666133-03.

There will be no calibrations for the old injector, Part Number 3867761 and Part Number 3867762, with compact pump or the new injector, Part Number 4010029, with the P10 pump. A new software calibration is required for a new injector with compact pump.

For troubleshooting procedures, refer to the Troubleshooting and Repair Manual, QSK Fuel System, Bulletin 3666113.

⚠ CAUTION ⚠

Both of the new fuel system components are not interchangeable with prior fuel system components. This is due to the design of the components and the software calibrations. The engine and OEM wiring harnesses installed before January 1998 also are not compatible with the new parts.

Introduction Date

The new fuel system components' effective date is as follows:

- **Engine Serial Number First 37181559**
- **Date of change was January 20, 1998**

Table 3, New Components

Component	Part Number	
Injector	4010029	
Compact pump	3331265	Without fittings
	4009869	With 90-degree inlet fitting, Part Number 3328590
	4009905	With straight inlet fitting, Part Number 3081957

Table 3, New Components		
Component	Part Number	
Injector copper seal (split in tangs)	3867687	Standard thickness
	3347933	0.010 oversize
	3347934	0.020 oversize
	3347935	0.030 oversize
Clamp	3867350	

Table 4, Old Components		
Component	Part Number	
Injector	3867761	
Injector	3867762	
P10 fuel pump	3088415	
Injector copper seal (split in tangs)	207244	Standard thickness
	3001658	0.010 oversize
	3001659	0.020 oversize
	3001600	0.030 oversize
Clamp	3867350	

The color coding and the part number of the o-rings on the old and the new injectors are as follows:

Table 5, O-Rings		
Location of O-Ring	Part Number	Color
Upper timing	3347939	Black
Lower timing	3347938	White
Upper rail	3347937	Green

Table 5, O-Rings

Location of O-Ring	Part Number	Color
Lower rail	3867651	Black

Note : The white and green o-rings have a coating that will come off during normal use. The loss of the coating will not alter the o-rings' function.

⚠ CAUTION ⚠

The increased capacity trapped volume spill injectors (new) are not interchangeable with the old injectors. Do not mix the new injector and old injector on the same engine. If the new injector is used to replace the old injector, the fuel pump, software calibration, and CPL must all be changed. In addition, if the engine was built before January 1998, both the engine harnesses and brackets must be changed.

In order to print this table properly, please set your printer paper orientation to "LANDSCAPE".

Table 6, Fuel Ratings

P10 and Old Injector Fuel Rating	Compact Pump and New Injector Fuel Rating
FR*4235 525 hp @ 2100 rpm	FR*4296 525 hp @ 2100 rpm
FR4270 700 hp @ 2100 rpm	FR4297 700 hp @ 2100 rpm
FR4283 490 hp @ 2000 rpm	FR4298 490 hp @ 2000 rpm
FR4237 600 hp @ 1800 rpm	FR4299 600 hp @ 1800 rpm
FR4236 500 hp @ 1800 rpm	FR4300 500 hp @ 1800 rpm
FR4231 480 hp @ 1800 rpm	FR4301 480 hp @ 1800 rpm
FR4205 600 hp @ 2100 rpm	FR4302 600 hp @ 2100 rpm
FR4254 525 hp @ 1800 rpm	FR4303 525 hp @ 1800 rpm
FR4227 450 hp @ 2000 rpm	FR4304 450 hp @ 2000 rpm
FR4261 510 hp @ 1800 rpm	FR4305 510 hp @ 1800 rpm
FR4367 660 hp @ 1800 rpm	FR4392 660 hp @ 1800 rpm

* FR = Fuel Rating

In order to print this table properly, please set your printer paper orientation to "LANDSCAPE".

Table 7, Control Parts List (CPL)

P10 and Old Injector CPL	Compact Pump and New Injector CPL
CPL 2008 JWAC	CPL 2810 JWAC
CPL 2225 JWAC/WA600	CPL 2560 JWAC/WA600
PP* 2008 JWAC	PP 2810 JWAC

Table 7, Control Parts List (CPL)	
P10 and Old Injector CPL	Compact Pump and New Injector CPL
PP 2225 JWAC/WA600	PP 2560 JWAC/WA600
CPL 2090 LTA/Marine	CPL 8193 LTA/Marine
* PP = Performance Parts	

Document History

Date	Details
2011-7-12	Module Created
2012-1-17	Adding in missing Marine CPL\'s and FR\'s